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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

October 2019 Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-2-Oct-2019-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Activity 2: Ice

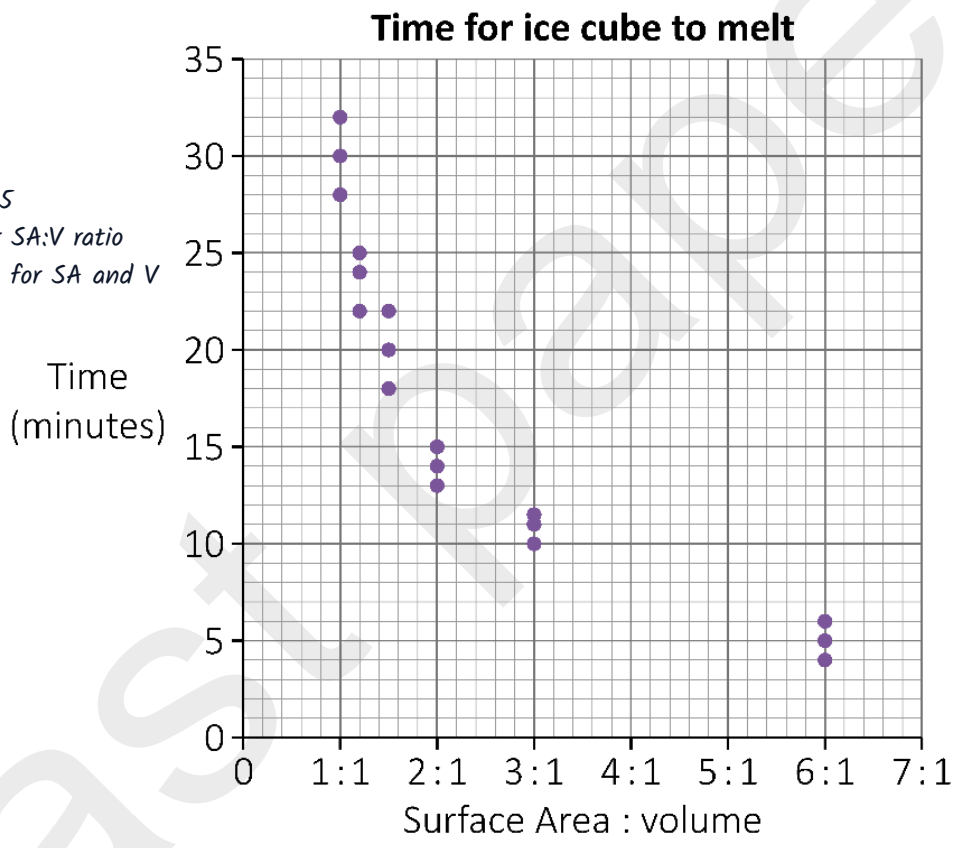
2 (a) Tom is studying science at college.
He is learning about ice.

Tom does an experiment.
He shows that the ratio of an ice cube's surface area to its volume affects the time it takes for the ice cube to melt.

Q2(a) Worked Solution:

This scatter diagram shows his results:

See below
1
37.5
1
15.625
1
2.4 : 1 or 12 : 5
Expresses their SA:V ratio
FT their values for SA and V
1
[4 marks]



past paper

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Tom has another ice cube.
Each side is 2.5 cm long.

Estimate how long this ice cube will take to melt.

[4 marks]

Your answer: **minutes**

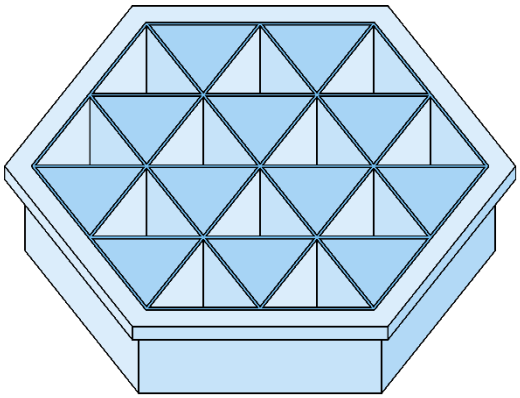
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2 (b) In the science lab, Tom finds this mould for making 24 triangular prisms of ice.

Q2(b) Worked Solution:
Any triangular prism correctly drawn
[1 mark]



Mould

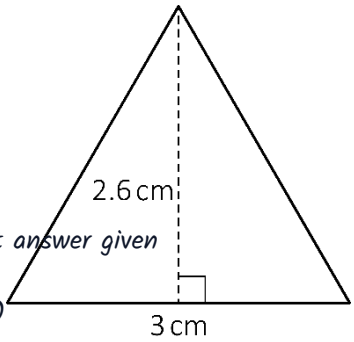
Draw a triangular prism.

[1 mark]

past paper

Examiner use only

2 (c) The triangular faces of each prism have base 3 cm and height 2.6 cm



Not drawn accurately

Q2(c) Worked Solution:
340.7 (cm3)
Award 5 marks if correct answer given
7
Level 2 – P001258 (v1.1)
1

Tom fills the hexagonal mould with water to a depth of 3.5 cm

Tom knows that:

- to calculate the volume of each triangular prism, he can use the formula

0.5 × 3 × 2.6 or 3.9 Volume = area of triangular face × depth of prism

- when a volume of water freezes, the ice that is formed has a volume 4% more than the volume of the water.

What will be the **total** volume of ice formed when all the water in the hexagonal mould freezes?

Calculates area of triangle
1
their 3.9 × 3.5 or 13.65
Calculates volume of water in triangular
[5 marks]

[5 marks]

Your answer: **cm³**

Please turn over

past paper

Examiner use only

2 (d) Icebergs are large lumps of ice that float in the sea.



Tom reads that the icebergs float because the density of ice is less than the density of seawater.

Q2(d) Worked Solution:

The density of ice is 0.92 g per ml and the density of the sea is 1.03 g per ml

89 (%)

Calculate 0.92 as a percentage of 1.03

Give your answer as a percentage to the nearest whole number.

Award 2 marks if correct answer given **[2 marks]**

1

0.92 ÷ 1.03 or 0.89(3...) or 89(.3...)(%)

1

89 (%)

[2 marks]

Your answer: **%**

2 (e) Tom reads that scientists are planning to move an iceberg across the sea to provide water to a very dry country.

The iceberg weighs 40 million tonnes.

Write 40 million using numbers.

[1 mark]

Q2(e) Worked Solution:

40 000 000

[1 mark]

Your answer:

past paper

Examiner use only

2 (f) The iceberg will melt as it travels.

By the time it arrives, the iceberg will weigh only 25 million tonnes.

By what fraction will the original iceberg have reduced in weight by the time it arrives?

Q2(f) Worked Solution:

3
8

Give your answer in its simplest form.

[2 marks]

Award 2 marks if correct answer given

1
15
40
1
3
8
8

[2 marks]

Your answer:

[Total marks: 15]

past paper

Please turn over

Examiner use only

Activity 3: Wages

3 (a) Faisal starts a new job.

He will work 37 hours a week for 48 weeks a year and is paid £8.25 an hour.

He must pay:

- 0% income tax on the first £12 500 he earns each year
- 20% income tax on anything over £12 500 that he earns.

His income tax payments are spread evenly over the 48 weeks.

Q3(a) Worked Solution:

Faisal's first weekly wage slip shows that he has paid £8.96 income tax.

He thinks he has paid more than he should.

See below

1 Is he correct?

Show how you decide.

[4 marks]

(37 × 48 × 8.25) – 12500 or
14652 – 12500 or 2152
Calculates taxable amount per year
1
0.2 × their 2152 or 430.4
or 2 × (their 2152 ÷ 10)
OE full correct method for finding 20%
"Their" is value calculated in 1st mark
[4 marks]

Your answer:

past paper

Examiner use only

3 (b) Sonya reads an article about the 5 : 3 : 2 rule.

The rule says you should spend your wages after tax in the ratio 5 : 3 : 2 on **needs : wants : savings**

- **needs** are bills like rent that must be paid
- **wants** are extra things like social life and new clothes which are not needed
- **savings** are amounts of money kept for the future.

Q3(b) Worked Solution:

Sonya earns £480 a week after tax.

Last week she spent £185 on going out with friends.

See below

1
3
10
or
0.3
or 480 ÷ 10 or 48
1
144 or 0.38(...) or 0.39 and no
Accept comparison of 38.5(%) or 39(%)
[2 marks]

Was this in line with the 5 : 3 : 2 rule?

Show how you decide.

[2 marks]

Your answer:

past paper

Please turn over

Examiner use only

3 (c) Dec needs to pay £908.64 each month for his mortgage and other bills.

This is 45% of his monthly income.

What is Dec's monthly income?

Give your answer to the **nearest ten pounds**.

[2 marks]

Q3(c) Worked Solution:

(£)2020

Award 2 marks if correct answer given

1

908.64 ÷ 0.45 or 2019.2(0)

May be seen in stages e.g. ÷ 45 then ×

100

9

Level 2 – P001258 (v1.1)

1

(£)2020

[2 marks]

Your answer: **£**

past paper

Examiner use only

3 (d) The manager of a small business claims that the mean salary in the company has gone up by over 10% in the last year.

One of the workers says that the gap between the highest and lowest earners has more than doubled.

Use this data about the company salaries to decide whether **each** of the statements is true. Show your working.

[5 marks]

Q3(d) Worked Solution:

See below

1

Annual salaries (£s)							
Last year (12 employees)				**This year (14 employees)**			
15 275	42 122	15 679		15 500	33 775	17 995	34 065
20 000	25 344	25 879		21 339	20 112	24 995	23 115
85 682	15 998	53 245		157 250	40 631	45 888	
55 333	70 507	32 196		60 200	80 555	17 160	

457 260 ÷ 12 = 38105

Correct mean for last year

H2S

1

592 578 ÷ 14 = 42327

Correct mean for this year

H2S

1

[5 marks]

Your answer:

Please turn over

past paper

Examiner use only

3 (e) Dave is a salesman.

He is paid a basic salary plus a bonus based on the sales he makes.

Last year Dave's bonus was $\frac{3}{8}$ of his salary.

Write $\frac{3}{8}$ as a decimal **and** as a percentage.

[2 marks]

Q3(e) Worked Solution:

Decimal:

0.375 and 37.5(%)
Award 2 marks if correct answer given
1
Decimal: 0.375
1

Percentage:

Percentage: 37.5(%)
Activity 4:
Archaeology
(Calculator Test)
Marks UPS
[2 marks]

[Total marks: 15]

past paper

Examiner use only

Activity 4: Archaeology

4 (a) Archaeologists study human history by digging up and analysing items from the past.

Chen is a volunteer with a community archaeology group.
At the weekends she helps to dig at an archaeological site.

The volunteers have to record exactly where each item is found.
They do this by using a coordinate grid laid over a map of the site.

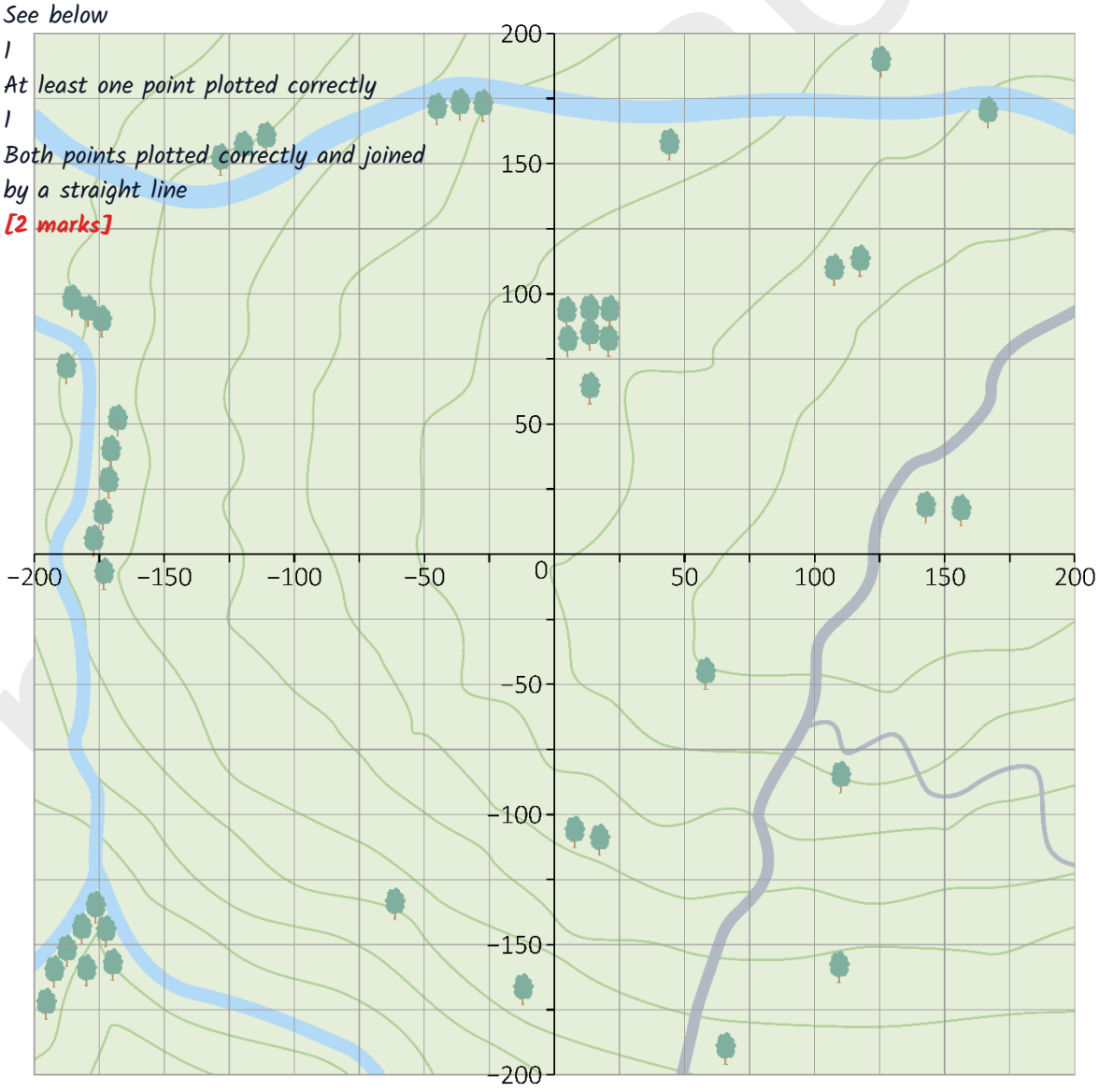
The volunteers have found the wall of a building.
It runs in a straight line from (–125, 100) to (150, –75)

Draw the position of the wall on the grid.

[2 marks]

Q4(a) Worked Solution:

See below
I
At least one point plotted correctly
I
Both points plotted correctly and joined by a straight line
[2 marks]

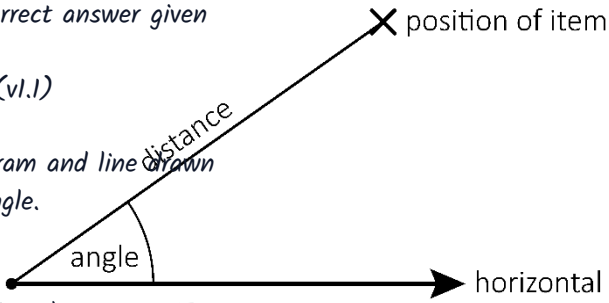


Examiner use only

4 (b) There is another way that the volunteers can record the position of an object:

- measure the object's distance from a fixed point and
- measure the angle (see example in the diagram below).

Q4(b) Worked Solution:
45 (°)
Award 2 marks if correct answer given
10
Level 2 – P001258 (v1.1)
1
Point drawn on diagram and line drawn to create required angle.
1
45 (°)
Implies first mark. No tolerance or FT as
[2 marks]



The volunteers find an object at the point (100, 100) on the coordinate grid.

i) Mark the point on the grid in **4 (a)**.

The volunteers measure the distance of the object from the fixed point (0, 0) They then measure the angle.

ii) What size is the angle?

[2 marks]

Your answer:	**degrees**

past paper

Please turn over

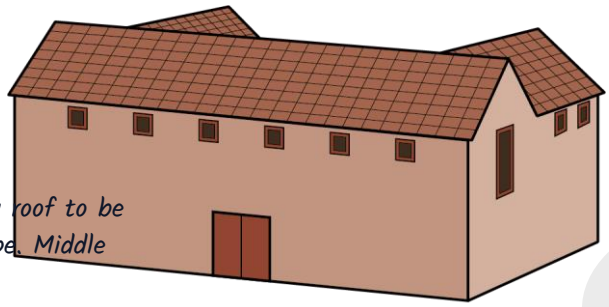
Examiner use only

4 (c) The volunteers find parts of a building.

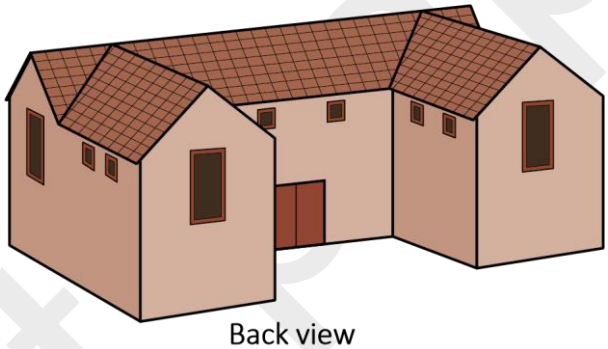
They think the building used to look like this:

Q4(c) Worked Solution:

See below
I
Shape of plan view correct (any orientation)
I
Shape of side elevation correct (ignore windows)
Ignore windows if drawn. Allow roof to be lower but must be correct shape. Middle vertical line may be missing
[2 marks]



Front view



Back view

pastpaper

Examiner use only

Draw:

- the plan view **and**
- the side elevation of the building.

[2 marks]

Plan view:

Side elevation:

past paper

Please turn over

Examiner use only

4 (d) Chen reads about some footprints that were preserved in volcanic ash.

The foot length was 18.8 cm and the stride length (the distance between the footprints) was 164.5 cm

Archaeologists use the formula: $R = \frac{\text{stride length}}{4 \times \text{foot length}}$

Chen uses a calculator to calculate R for the footprints found in the volcanic ash.

She enters 164.5 ÷ 4 × 18.8 and gets the answer 773.15

Explain what Chen has done wrong.

[1 mark]

Q4(d) Worked Solution:

Any correct explanation relating to order of operations
e.g. she should have multiplied first
You have to do 4 × 18.8 first
[1 mark]

past paper

Examiner use only

4 (e) If R is less than 2.0 the person was probably walking.
If R is greater than 2.9 the person was probably running.
If R is in between 2.0 and 2.9 it is not possible to decide.

Tick the correct statement about the person whose footprints were preserved in the volcanic ash.

Q4(e) Worked Solution:

☐ They were probably walking.

☐ They were probably running.

'It is not possible to decide' with 2.18(75)

☐ It is not possible to decide.

Award 2 marks if correct answer given

1

Explain your choice.

2.18(75) **[2 marks]**

Accept rounding to 2.2 or 2.19

1

'It is not possible to decide' with 2.18(75)

Compares 2.18(75) or 2.2 or 2.19 with 2.9 (and 2.0)

[2 marks]

past paper

Examiner use only

4 (f) There are 10 volunteers in the community archaeology group.

The probability that they all attend on a Saturday is 0.35
The probability that they all attend on a Sunday is 0.12

The site is open next weekend.

Q4(f) Worked Solution:

Calculate the probability that all the volunteers will attend on both Saturday and Sunday next weekend.

Give your answer as a percentage.

[2 marks]

4.2 (%)
Award 2 marks if correct answer given
1
0.35 × 0.12 or 0.042
H26
1
4.2 (%)
H27
[2 marks]

Your answer: **%**

past paper

Examiner use only

4 (g) The site was open on Saturday and Sunday for 26 weekends during last year.

This table shows the number of days each volunteer attends out of the 52 days available.

The information for Chen is missing.

Q4(g) Worked Solution:

No with correct explanation

Volunteer	Number of days
Ada	48
Barry	25
Tim	30
Chris	24
Gwen	43
Sia	39
Rita	40
Rav	18
Jon	35
Chen	

Chen says that the mode of the 9 values is 48

Is she correct?

Explain your answer.

[1 mark]

past paper

Please turn over

Question 4(g) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

e.g. there is no mode

e.g. 48 is the highest not the mode

H23b

[1 mark]

Examiner use only

4 (h) The median number of days for the 10 volunteers including Chen was 33.5

What fraction of the days did Chen attend?

Give the fraction in its simplest form.

[3 marks]

Q4(h) Worked Solution:
8
13
Award 3 marks if correct answer given
1
35
Median for 9 volunteers i.e. current 5th value
H23a
1
32
[3 marks]

Your answer:

[Total marks: 15]

This is the end of the external assessment.

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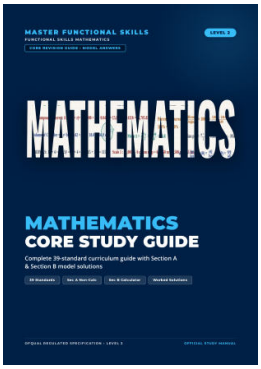
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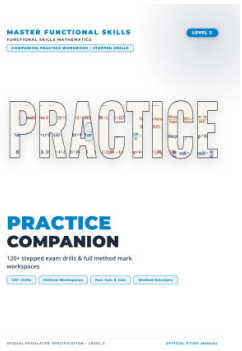


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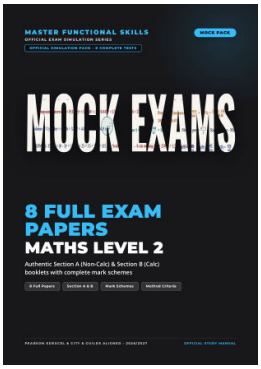


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